

Quenverell Computational Ltd, a Tarnovell company

Computational analysis for decisions in discovery

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Edinburgh, UK

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Scientific enquiries: Tomasz Zawiernicki or Fiona Macrae

Engagement coordination: Julie Lindsey

1. Start with the question

A computational output is useful when its relationship to a biological decision is clear. Quenverell begins with the question, the source of the data and the decision the analysis is expected to support. We establish what the material can tell us before choosing a model. When a dataset cannot separate the effect of interest from its preparation history, we say so; when a virtual ranking depends on uncertain assumptions, those assumptions remain visible in the recommendation.

Our bioinformatics services interpret expression data, integrate evidence for target selection and revisit existing sequencing material. Our computational chemistry services prepare virtual libraries and organise docking candidates for experimental follow-up. These services can be combined in an engagement, but their outputs answer different questions. Differential expression is not a target-validation experiment, and a docking score is not a measured affinity.

- Define the biological question and the intended decision in a target card.
- Establish provenance and resolve identity questions at data acceptance.
- Agree the model gates before committing to a final interpretation.
- Deliver a technical note with the analysis explanation and its limitations.
- Retain an analysis notebook that connects the recommendation to the source material.

This catalogue describes the service boundaries current at 12/06/2024. A written scope establishes the inputs, analysis population and delivery requirements for each engagement. The scope also identifies dependencies that could prevent the requested comparison or recommendation. No catalogue entry substitutes for that assessment.

2. Choosing the service

Service selection by decision

Question	Service	Principal output
Which expression features differ between defined conditions?	RNA-seq differential expression	Named contrasts with log2 fold change and adjusted probabilities
Which targets merit further biological investigation?	Omics target prioritisation	Evidence-based target cards and a qualified candidate set
Can existing sequencing material answer a revised question?	NGS reanalysis	Revised analysis with comparison to the earlier interpretation
Which library entries should enter a modelling campaign?	Virtual library triage	Curated candidate set with provenance and exclusion reasons
Which modelled candidates deserve experimental follow-up?	Docking and hit clustering	Ranked and clustered candidates with interpretation limits

Choose RNA-seq differential expression when the sample design supports a defined comparison and the immediate need is an expression contrast. Choose Omics target prioritisation when the task is to integrate that contrast with other evidence and recommend biological follow-up. NGS reanalysis is appropriate when existing sequence material or count data require a new interpretation, reference or analysis question. It begins with a feasibility assessment of what has been retained.

Virtual library triage establishes a usable modelling population and addresses representation, duplication and diversity. Docking and hit clustering adds a target-dependent model ranking and representative selection. A client with an already curated library may require the latter service alone, provided the preparation and provenance can be accepted. A combined campaign keeps the curation decision separate from the model preference.

Tomasz Zawiernicki coordinates bioinformatics scoping. Fiona Macrae coordinates computational chemistry scoping. Julie Lindsey aligns the agreed scientific scope with delivery arrangements. Unclear service boundaries are resolved in the scope rather than left for the final report.

3. RNA-seq differential expression

This service estimates expression differences between named conditions using an agreed design. The question may concern a treatment, a phenotype or another recorded condition, but the available biological replication and covariates must support that comparison. We assess sample mapping and technical grouping before fitting the model. A comparison that is inseparable from a preparation group is reported as a limitation rather than presented as an independent biological effect.

Input and delivery requirements

Required inputs	Deliverables
Sequence data or an interpretable count matrix	Accepted input inventory and metadata reconciliation
Sample identities, conditions and relevant covariates	Documented design and named contrasts
Preparation and sequencing history where available	Diagnostic review of sample relationships and technical grouping
Reference and annotation information	Contrast tables and reproducibility appendix

Contrast tables report log2 fold change with a stated direction and adjusted probabilities with the adjustment method identified. Filtered, missing or non-estimable entries remain distinguishable from tested entries. The technical note explains the analysed population, the model and material uncertainty. The analysis notebook preserves the relationship between source files, sample metadata and release tables.

The output supports comparison and hypothesis formation. It does not establish causation or show that changing a differentially expressed feature will change the phenotype. Expression results must be interpreted in the context of the material sampled. Technical diagnostics cannot supply missing clinical or experimental metadata, and an attractive expression pattern cannot prove sample identity.

Published 2024 list price: GBP 12,500 per dataset; no volume discount. Dataset boundaries, upstream processing requirements and the intended contrasts are confirmed in the written scope.

4. Omics target prioritisation

Omics target prioritisation asks which biological targets deserve focused follow-up, given the evidence available and the intended intervention. We first define what makes a target relevant to the question. Expression evidence may contribute, but it is considered alongside the provenance, independence and biological meaning of other evidence streams. A repeated observation from the same source is not counted as independent support.

- Required: the biological question, accepted omics outputs and the population or material they represent.
- Required: source annotations and the provenance of any additional evidence used for ranking.
- Required: the proposed intervention context and known constraints on follow-up.
- Delivered: target cards that separate observed evidence, interpretation and unresolved questions.
- Delivered: a prioritised candidate set with reasons for progression, deferral or exclusion.
- Delivered: a technical note and analysis notebook describing integration and sensitivity to assumptions.

The model gate agrees how evidence will be weighed before the final candidate set is selected. We identify disagreements between sources and explain where an apparently strong priority depends on a particular assumption. Where the evidence does not support a strict ordering, the output may distinguish groups for follow-up rather than impose an artificial rank.

Prioritisation is a decision aid, not experimental target validation. Association, expression change and plausible mechanism have different evidential strength. Missing evidence does not automatically demonstrate absence of a biological role. The delivery identifies experiments or source clarification that could reduce the uncertainty affecting selection.

This service is quoted to scope. The quotation depends on the evidence to be integrated, its readiness for analysis and the form of the target cards required. No fixed list price is published in this catalogue.

5. NGS reanalysis

NGS reanalysis revisits retained sequencing material to address a revised biological question, an updated interpretation or an identified limitation in earlier processing. It begins by establishing what survives: sequence files, processed data, reference information, sample metadata and the earlier analysis explanation. We distinguish a rerun of an existing method from an analysis that changes the population, reference or question.

Feasibility depends on the retained material

Input assessment	Delivery consequence
Original sequence material and metadata available	Upstream choices can be reconsidered within scope
Only processed data retained	Analysis is limited by the transformations already applied
Earlier scripts and reference versions retained	A direct comparison of analysis choices may be possible
Earlier provenance incomplete	The technical note identifies what cannot be reconstructed

Required inputs include the revised question, source identifiers and the earlier report or delivery where available. We reconcile the retained files with the metadata and define the comparison between earlier and revised outputs. Changes in annotation, feature handling, filtering or model design are documented so that altered conclusions can be traced to their cause.

Deliverables include an accepted input inventory, the reanalysis outputs, a technical note describing substantive differences and a reproducibility appendix. Where the original analysis cannot be reconstructed, we explain that boundary. We do not claim to have reproduced the earlier result merely because a new analysis produces a similar conclusion.

Reanalysis cannot recover unrecorded sample identity or create biological replication absent from the original design. A revised reference may alter feature assignment without resolving the underlying biological uncertainty. The service is quoted individually after review of the retained material and the requested question.

6. Virtual library triage

Virtual library triage prepares a defensible candidate set for a target campaign. The starting point is the provenance and meaning of each library entry: what structure is represented, which source identifier accompanies it and whether its representation is suitable for the intended modelling. We retain the source relationship while resolving duplicates, ambiguous structures and preparation choices.

- Required: a structure library with stable source identifiers and its origin.
- Required: the target-campaign question and any stated purchase or chemistry constraints.
- Required: availability information where procurement is part of the intended decision.
- Delivered: a curated candidate set and a preparation summary.
- Delivered: an exclusion and query log that preserves the original identifiers.
- Delivered: a diversity assessment and technical note explaining the triage rules.

The triage rules distinguish an uninterpretable entry from an interpretable entry that falls outside the agreed selection constraints. Salts, mixtures, stereochemical ambiguity and relevant molecular states are handled explicitly. Diversity is considered so that a densely represented chemotype does not occupy the candidate set simply because the source library contains many similar entries.

The service does not, by itself, produce a target-dependent docking ranking or establish biological activity. Availability is a dated source statement and may need confirmation at purchase. A curated virtual entry does not demonstrate the identity or quality of physical material supplied later. Those questions belong to the experimental and procurement workflow.

Published 2024 list price: GBP 9,500 per target campaign; no volume discount. The library boundary, required preparation and selection constraints are agreed in the scope. Docking and hit clustering is a distinct service and is included only when expressly scoped.

7. Docking and hit clustering

Docking and hit clustering organises an accepted candidate set around a target-dependent computational model. The biological question must be clear enough to judge whether the target information and modelling assumptions are relevant. We inspect the provenance of the target material and the prepared library before treating a rank as useful evidence for selection.

Docking campaign requirements

Required inputs	Deliverables
Accepted structures and source identifiers	Prepared-input and assumption register
Target information with provenance	Model description and limitations
Selection objective and practical constraints	Ranked candidates and cluster representatives
Any relevant experimental evidence	Technical note linking model preference to follow-up

Docking scores are reported in kcal/mol where applicable, with the scoring procedure identified. They are interpreted within that procedure and target model. We examine candidate poses and chemical plausibility alongside rank, and retain the reason for choosing a cluster representative. A diversity-based substitution is described as such rather than presented as the top-scoring entry.

The delivery can include a model-sensitivity assessment when alternative defensible preparation choices could affect the recommendation. Entries with unresolved identity or unsupported states remain flagged. The analysis notebook records preparation, model settings and selection decisions so that later experimental results can be connected to the original recommendation.

A favourable score is not a measured affinity, and a plausible pose is not proof of binding. Ranking cannot demonstrate selectivity, activity in a biological system or suitability for development. This service is quoted to scope, including the target information available, model alternatives and intended selection output.

8. Combining services without losing the evidence trail

An engagement may move from RNA-seq differential expression to Omics target prioritisation, or from Virtual library triage to Docking and hit clustering. Each transition is a model gate. The input accepted for the next service is identified, and any limitation from the preceding service travels with it. A later integration step does not remove uncertainty in the original observations.

Examples of service handover decisions

Service connection	Gate question
RNA-seq differential expression to Omics target prioritisation	Are the contrasts interpretable for the intended target question?
NGS reanalysis to RNA-seq differential expression	Does the retained material support the agreed design?
Virtual library triage to Docking and hit clustering	Are the representations and source mappings suitable for the model?
Docking and hit clustering to experimental selection	Which recommendations remain robust enough to justify follow-up?

A combined technical note distinguishes the observed input, the modelled interpretation and the proposed action. We avoid a single composite rank that hides incompatible evidence or missing provenance. Where different evidence streams disagree, the disagreement is retained and its practical consequence explained. The recipient can then see whether a candidate is favoured by direct observations, model assumptions or a deliberate diversity choice.

Combined scopes define responsibility for resolving queries and approving each gate. If an upstream issue blocks the intended next step, the engagement pauses that interpretation and considers a narrower deliverable. Work can remain useful without implying that every originally proposed conclusion is supportable.

9. Data acceptance and confidential transfer

Every received data drop is registered before analysis. We identify the sender, receipt date, declared version and relationship to earlier material. Original files are retained separately from transformed inputs. Checksums and file inventories link the analysis to the material actually received. Nadia Rahmani coordinates data curation and the reproducibility trail.

- Agree the confidential transfer route and authorised recipient before sending material.
- Supply metadata with the files rather than relying on folder names to carry identity.
- Identify replacement exports explicitly and explain what changed.
- Include the provenance of references, annotations and externally processed outputs.
- Resolve discrepancies through documented correspondence before affected modelling proceeds.

Data acceptance is a scientific and provenance assessment as well as a technical check. A readable file may still have uncertain sample identity, an unclear count definition or an unsupported molecular representation. We raise such issues early and explain what clarification would permit progression. Missing information is recorded rather than replaced with an undocumented guess.

Client material and engagement outputs are treated as confidential. Access is limited to staff supporting the agreed work. Transfer arrangements and retention expectations are established in the written scope. Delivery recipients are confirmed before release; a person coordinating the quotation is not automatically assumed to be the scientific recipient.

The acceptance trail distinguishes received, held, accepted and rejected inputs, including acceptance with an authorised limitation. This keeps the final technical note aligned with the material that entered the model and makes later reanalysis easier to scope.

10. Reproducible delivery and interpretation

A standard delivery contains the agreed analysis outputs, a technical note and a reproducibility appendix. Tables retain source identifiers and define their columns. The technical note states the question, accepted inputs, methods, decisions and material limitations. The appendix identifies analysis versions and choices that would be needed to reconstruct the delivered interpretation.

Delivery content agreed during scoping

Delivery component	What the recipient should be able to establish
Input inventory	Which source material was analysed
Acceptance summary	Which queries and limitations affected the analysis
Results or selection files	What was estimated or recommended, with clear identifiers
Technical note	Why the output supports the stated decision
Reproducibility appendix	How the output relates to the executable analysis

An independent colleague checks the relationship between the analysis and the released explanation. For expression contrasts this includes sample mapping, direction and adjusted-probability definitions. For a virtual candidate set it includes source mapping, preparation flags and the rationale for representative selection. The check supports release control; it does not turn a computational prediction into experimental proof.

We describe uncertainty where it affects a decision. This may mean presenting an alternative model, qualifying a ranking or stating that a contrast cannot be separated from technical grouping. The recipient receives the limitation with the output rather than having to discover it in an internal analysis notebook. Follow-up questions can then focus on the evidence most likely to change the decision.

11. Published prices and written scope

Published list prices for 2024

Service	Unit	2024 list price	Volume discount
Virtual library triage	Per target campaign	GBP 9,500	None
RNA-seq differential expression	Per dataset	GBP 12,500	None

Omics target prioritisation, NGS reanalysis and Docking and hit clustering are quoted individually. Their scope depends on the question, the readiness and provenance of the inputs and the interpretation required. A quotation identifies the work included, the deliverables and any dependencies on further information. The published list entries do not imply that these additional services are included.

Julie Lindsey coordinates the engagement scope and delivery arrangements. Scientific feasibility is considered before a schedule is agreed. A late replacement data drop, a changed biological question or a newly requested model can affect the agreed delivery. We describe that consequence and obtain agreement to the revised scope before extending the analysis.

- Define the dataset or target-campaign boundary.
- Identify the supplied inputs and outstanding clarification.
- State which service outputs are included.
- Agree model gates, recipients and delivery formats.
- Record how changes to inputs or interpretation will be handled.

This catalogue does not publish a standard completion interval. The written scope provides the engagement schedule after input review. Procurement, experimental testing and physical sample handling are included only if a separate written arrangement expressly provides for them. Computational selection alone does not authorise purchase or testing.

12. Preparing an enquiry

Send a concise statement of the biological question and the decision you expect to make from the output. Describe the material available, its origin and the metadata or identifiers that connect it to the question. If an earlier analysis exists, include its explanation and say what you wish to reconsider. If the work concerns virtual candidates, describe the target context and whether the intended selection is for modelling, purchase or experimental follow-up.

Initial enquiry checklist

Enquiry topic	Information that helps scoping
Expression comparison	Conditions, biological units, preparation history and intended contrasts
Target prioritisation	Evidence sources, intervention context and follow-up constraints
Reanalysis	Retained files, earlier methods and revised question
Library triage	Structures, source identifiers and selection constraints
Docking campaign	Target provenance, prepared candidates and interpretation objective

Tomasz Zawiernicki, Head of Bioinformatics, is the scientific contact for RNA-seq differential expression, Omics target prioritisation and NGS reanalysis. Fiona Macrae, Computational Chemistry Lead, is the scientific contact for Virtual library triage and Docking and hit clustering. Nadia Rahmani supports provenance and reproducibility discussions. Julie Lindsey, Project Manager, coordinates the written scope and delivery arrangements.

Where the evidence is not ready for the proposed analysis, the first recommendation may be to resolve an identity question, obtain metadata or revise the comparison. That is part of making the engagement useful. We aim to leave the client with an interpretable output, a clear basis for the next decision and an analysis trail that can be revisited when new evidence arrives.

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